

WASP-12b

R-1958

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151215 · created 2026-08-06T16:15:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457371.77188 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1185 +/- 0.0058
Transit depth (Rp/Rs)^2	1.4 +/- 0.14 [%]
Orbital Inclination (inc)	84.57 +/- 1.86
Airmass coefficient 1 (a1)	1.0255 +/- 0.007
Airmass coefficient 2 (a2)	-0.0222 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	#2 - [177, 157]
Optimal Aperture	4.25
Optimal Annulus	12.42
Transit Duration (day)	0.1262 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1185 ± 0.0058 vs 0.1178 (deviation 0.12σ)
Duration fit vs published	0.1262 d vs 0.125 d (deviation 0.38σ)
Mid-time O–C	0.5 min
Depth SNR	20.0
Residual scatter	0.68 %

Light curve

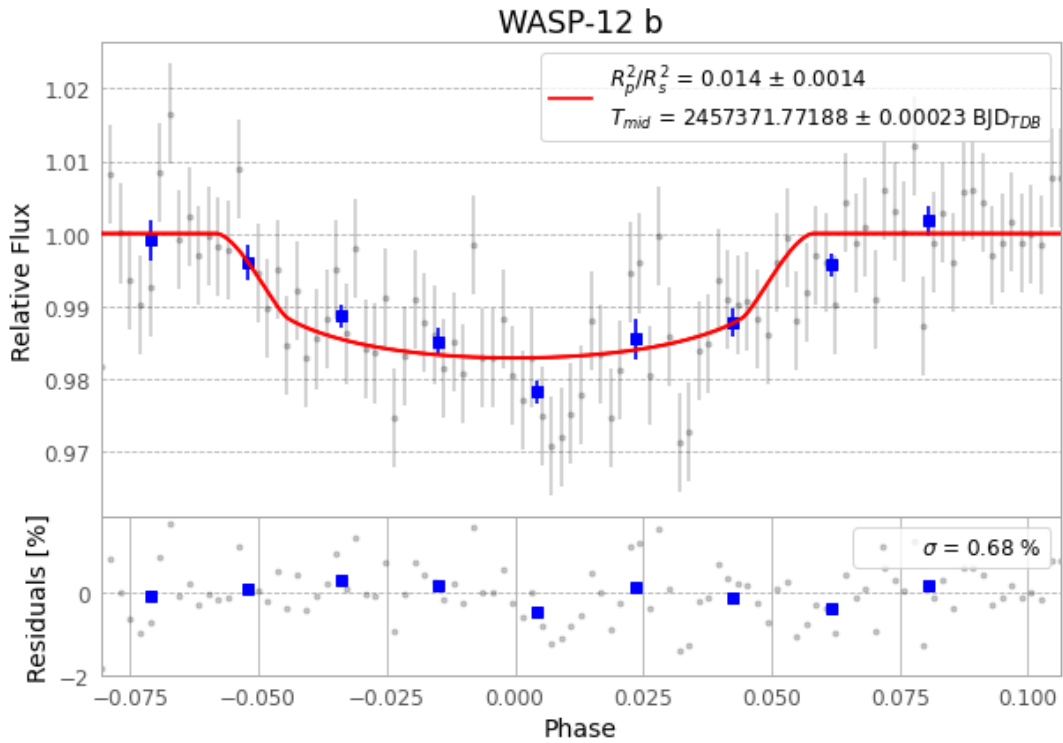


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-14.png (SHA-256 1061cb9986abb08a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4f4d909eafe0a0ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

99 FITS frames (65 MB), WASP-12151215041512.FITS → WASP-12151215090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151215.log (e86af7470b3c...), FinalLightCurve_WASP-12 b_2015-12-14.png (1061cb9986ab...), FinalLightCurve_WASP-12 b_2015-12-14.csv (da5a0d197aa9...)

Compute resources

Wall time 127.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 16:15Z.